

POLITECNICO
MILANO 1863

SCUOLA DI INGEGNERIA INDUSTRIALE
E DELL'INFORMAZIONE

Space Propulsion Project

Academic Year 2024/2025
Montemagno Group

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1. Introduction

In the past decade, there has been growing interest in small, low-cost satellite systems like PocketQubes and CubeSats. To meet mission requirements and extend their lifespan, these satellites often need micro-propulsion systems for orbit corrections or formation flying.

MEMS (Micro-Electro-Mechanical Systems) integrate miniaturized electro-mechanical components to create scaled-down technologies. The concept dates back to 1965 with the development of a resonant-gate transistor. Due to their significantly smaller scale, MEMS devices require special design and manufacturing considerations.

The development of miniaturized propulsion systems for space applications benefits from MEMS technology. Silva et al. [1] review the current status of MEMS-based electric and chemical propulsion systems, highlighting their performance capabilities.

1.1 Micro-propulsion systems

1.1.1 Micro-nozzles fundamentals

Micro-nozzles are key components in micro-electromechanical systems (MEMS) that will be referred to often during the study. MEMS-based micro-nozzles facilitate precise thrust generation necessary, for example, for orbital adjustments and attitude control in 3U CubeSats, as the one considered in this study. Design challenges include mitigating losses and optimizing nozzle geometry to enhance performance. During the sizing, the nozzle was designed for each considered propulsion system in order to best adhere to the required specifications in terms of expansion ratio, as well as pressure and temperature in different sections.

Throat section

For each specific design it was necessary then to characterize the throat conditions. The throat section has been designed obtaining pressure and temperature from the constrains imposed on the nozzle inlet conditions through isentropic calculations. The throat area was then evaluated through the following equation:

$$A_t = \frac{T}{P_0 \cdot C_T} \tag{1.1}$$

where C_T is the Thrust coefficient, defined as:

$$C_T = \lambda \cdot \sqrt{\frac{2\gamma^2}{\gamma - 1}\left(\frac{2}{\gamma + 1}\right)^{\frac{\gamma+1}{\gamma-1}}\left(1 - \left(\frac{P_e}{P_0}\right)^{\frac{\gamma-1}{\gamma}}\right)} + \frac{P_e - P_a}{P_0} \cdot \frac{A_e}{A_t} \tag{1.2}$$

The λ parameter was imposed on the thrust coefficient to include from the start of our analysis the 2D losses related to the divergent angle:

$$\lambda = \frac{1 + cos(\alpha)}{2} \tag{1.3}$$

Exit section

Once the throat section was fully characterized the next step involved the exit section, where also in this case temperature and pressure conditions where evaluated through isentropic calculations while the characteristic area was found through the previously imposed expansion ratio.

Inlet section

The last section that needed to be sized was the inlet, where the area was evaluated through the area ratio equation involving the mach number at the inlet:

$$\epsilon_c = \frac{1}{M}\sqrt{\left[\frac{2}{\gamma+1}\left(1+\frac{\gamma-1}{2}M^2\right)\right]^{\frac{\gamma+1}{\gamma-1}}} \tag{1.4}$$

Also in this case temperature and pressure were then evaluated though isentropic calculations.

Nozzle Length

A key sizing parameter was then the total length of the nozzle, obtained constraining the converging and diverging angles, which values are taken based on the needs of each configuration studied.

$$L_d = \frac{D_e - D_t}{2 \cdot tan(\alpha)} \qquad L_c = \frac{D_b - D_t}{2 \cdot tan(\beta)} \tag{1.5}$$

Mass Flow Rate and Exit Velocity

Apart from the dimensions of the nozzle, two other key characteristics can be obtained from this sizing procedure: the mass flow rate and the exit velocity. The mass flow rate is obtained from the thrust equation considering the 2D losses contribution as follows:

$$\dot{m} = \frac{T - (P_e - P_a)A_e}{\lambda \cdot u_e} = \frac{T}{I_{sp}g} \tag{1.6}$$

while the exit velocity is obtained having considered the exit pressure previously evaluated, using the following expression:

$$u_e = \sqrt{\frac{2\gamma}{\gamma-1}RT_0\left(1-\left(\frac{P_e}{P_0}\right)^{\frac{\gamma-1}{\gamma}}\right)} \tag{1.7}$$

1.1.2 Effect of Reynolds Number

Unlike large-scale nozzles where boundary layers are relatively small, micro-nozzles could reach very low Reynolds numbers at the throat area (Re_{throat} typically < 500). In these regimes, it becomes important to account for viscous effects. The viscous boundary layer can occupy a significant portion of the nozzle's cross-section, leading to performance uncertainty and losses compared to ideal isentropic expansion.

In an experiment done by [2], a nozzle with $d_{throat} = 280\mu m$ had been tested with different Re_{throat}, and it was found that the uncertainty increases significantly at low Re_{throat}. For example, at a Re of 51.3, the thrust had an uncertainty of 63%, while at Re of 414, this percentage decreased to 3.4%.

1.1.3 Effect of Expansion Ratio

Viscous effects are also influenced by the expansion ratio of the nozzle, depending on operating pressure and the ambient pressure, increasing the expansion ratio might increase or decrease the performance. Work in [3] quantitatively discusses how varying the expansion ratio affects the performance of a linear micro-nozzle. The study simulated nozzle performance for expansion ratios ranging from 3.6 to 11.4 at operating pressures of 1 kPa, 10 kPa, and

50 kPa. These pressures correspond to distinct flow regimes: transition, slip, and continuum, respectively, all evaluated against an ambient pressure of 1 Pa.

In the transitional flow regime, the performance, measured by thrust and specific impulse, was found to decrease as the expansion ratio increased. The paper attributes this trend to the dominance of viscous losses originating from the relatively thick subsonic layer near the nozzle wall at these low Reynolds numbers, which outweighs any potential gains from flow expansion.

Conversely, in the slip and continuum flow regimes, both thrust and specific impulse were observed to increase with rising expansion ratio, eventually stabilizing for ratios of 8.9 and higher. In these higher-pressure regimes, the positive effects of enhanced flow expansion become more dominant relative to the viscous losses, leading to improved performance with larger expansion ratios.

On the other hand, increasing the expansion ratio caused a decrease in the specific impulse efficiency $\eta_{I_{sp}}$. Specifically, comparing an expansion ratio of 3.6 to 11.4, the efficiency approximately dropped by 12.5% at 1 kPa, 8.6% at 10 kPa, and 5.6% at 50 kPa. It can be observed that the higher the operating pressure, the smaller the losses. This reduction in efficiency occurs because the larger surface area associated with higher expansion ratios leads to increased total viscous losses.

1.1.4 Effect of cross-sectional profile

Viscous effects also influence the choice of the geometric configuration of the nozzle's cross-section, impacting performance. Work in [3] compares the performance of a linear micro-nozzle (rectangular cross-profile) and an axisymmetric micro-nozzle (circular cross-profile). Both nozzles were designed with identical inlet, throat, and exit areas, as well as the same converging and diverging half-angles.

In general, it was found using 3D simulations that the axisymmetric nozzle demonstrated superior performance. At 10 kPa operating pressure, the axisymmetric nozzle produced ∼9.5% higher thrust compared to the linear nozzle. At 30 kPa operating pressure, the axisymmetric nozzle still showed better performance, with ∼3.6% higher thrust.

In the continuum regime (50 kPa operating pressure), the performance difference became negligible, with the axisymmetric thrust being only ∼1.6% higher and I_{sp} nearly the same. Furthermore, the study highlights a significant size advantage: the total volume of the axisymmetric nozzle (0.086 mm^3) was found to be approximately half that of the linear one (0.174 mm^3) for the same specifications for the same performance.

1.2 Design methodology

In the present work, we will propose and discuss the design of three different micro-propulsion systems, each based on a different technology. These are, in the order presented:

- Solid propellant motor
- Cold gas thruster
- Vaporizing liquid resistojet

Each propulsion system will be designed for a reference 3U Cubesat with fixed mass of 4 kg, providing a thrust of 5 mN, for a total ΔV of 0.5 $\mathrm{m\,s^{-1}}$. This corresponds to a cumulative burn time of 400 seconds. The modeling approach and sizing procedure for each system will be detailed further in each of the work's sections, as they differ appreciably in nature.

Besides the nominal sizing, an overview of each system will be presented by means of an schematic which details the design configuration. Furthermore, each solution will be analyzed in terms of how uncertainties in key parameters of the thruster design affect the final performance, by means of stochastic simulations. Finally, a comparison and critical analysis will be performed, contrasting propulsion performance parameters common among the systems, such as delivered thrust, specific impulse, minimum impulse bit and estimated total volume.

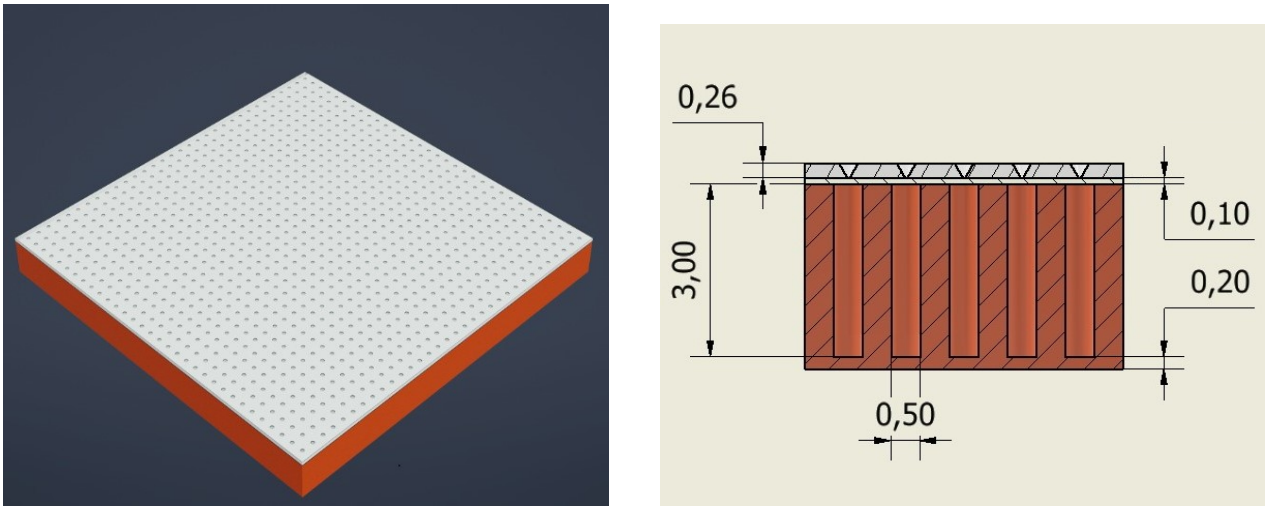
2. Solid Propellants

2.1 Configuration

The design of this propulsion subsystem is based on the solid propellant micro thruster developed in reference [4], although with some proper modifications: the first being the choice of a much lower combustion chamber pressure of 45 bar, as it seems to be more in line with other vacuum-based Solid Rocket Motor (SRM) designs found in the literature. The actual value was found by iterating the design in order to have rocket dimensions more in line with manufacturing procedures; and the second being the addition of a diverging nozzle. The nozzle consists only of the diverging part, as it is easier to manufacture than a full convergent-divergent one, while still yielding good performance.

The system is composed of a rectangular base plate where 1024 precision holes are drilled via conventional micro-machining, although it could also be made through additive manufacturing. Besides, a top plate is present where the same number of divergent-only nozzles are machined using Femtosecond Laser Machining (FLM). The material selected for the base plate was epoxy resin [4], while the material selected for the nozzle plate was silver as stated in [5]. The use of other metals could be explored but will be kept as a future development.

Each hole is filled with aluminized AP-HTPB propellant shaped in an end-burner configuration, which gives a constant burning area. The ignition system is very thin pcb inserted between the bottom and top plate. The final design is reported in figure 2.1.



(a) Full array

(b) Cross section of a few motors

Figure 2.1: SRM final design

2.2 Sizing

The workflow used for the sizing is illustrated in figure 2.2, and below the main sections are discussed.

2.2.1 Propellant Characterization

As stated previously, the propellant used is aluminized AP-HTPB, which was characterized previously by processing data from several BATES (Ballistic Test) motors. In that test, the burn time of test motors of known grain web thickness was measured at different operating pressures. From this, the coefficients of the Vieille's law and the characteristic velocity were obtained. The results are reported in table 2.1.

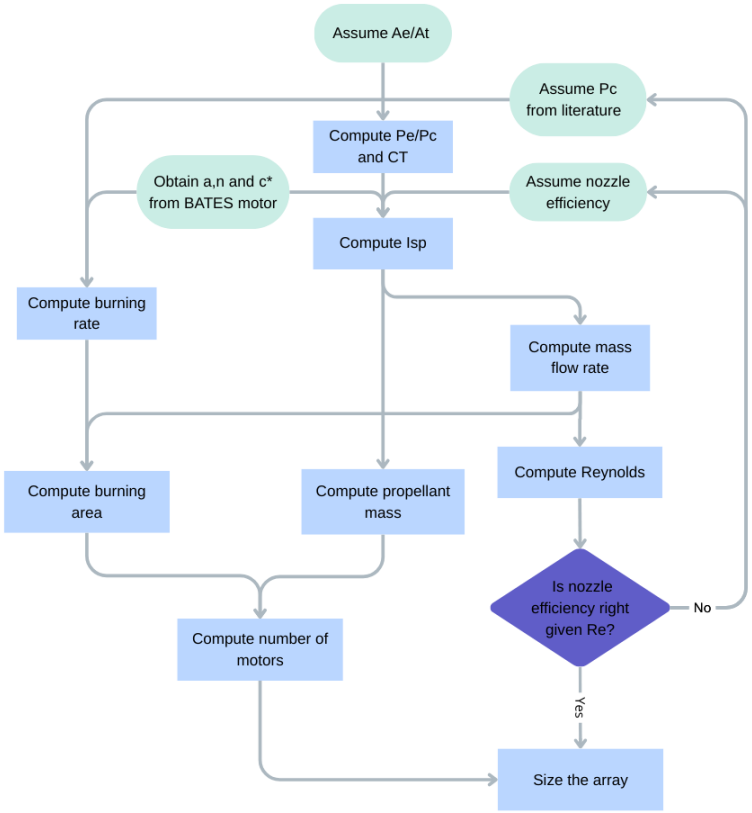


Figure 2.2: SRM sizing workflow

Table 2.1: Aluminized AP-HTPB parameters

n	$a\ [mm \cdot bar^{-n} \cdot s^{-1}]$	$c^*\ [m/s]$
0.3821 ± 0.0027	1.7270 ± 0.0184	1513 ± 14.8

Molar mass and specific heat capacity ratio of the combustion gas have been computed with the NASA CEA thermochemistry equilibrium code [6].

2.2.2 Nozzle sizing

Since the motor has to work in vacuum, an expansion ratio ϵ of 100 has been chosen, as it allows enough expansion of the gas while not occupying too much space. From ϵ, the design approach was to first solve numerically for the pressure ratio P_c/P_e, then use it to compute the thrust coefficient C_T using eq. 1.2, correcting it to account for the efficiency of the micro nozzle, assumed to be 70% based on the findings by Bruccoleri et al.[7]. Subsequent computation of the Reynold's number at the throat indicates that value of efficiency to be reasonable. Having obtained both c^* and C_T we can compute the specific impulse I_{sp} as:

$$I_{sp} = \frac{c^* C_T}{g_0} \tag{2.1}$$

which yields a value of 198.7 s. From that, the mass flow rate is computed with eq. 1.6, as well as the throat and exit diameters since the chamber pressure has been fixed. The resulting diameters are 33 μm for the throat and 330 μm for the exit section. The divergent nozzle

shape was chosen to be conical for ease of manufacturing, with a half angle α of 30°, found to be optimal by Louisos et al. [8]. Finally, eq. 1.5 leads to a diverging nozzle length of 257 μm.

2.2.3 Grain sizing

This information can now be used inside the rocket equation (given the ΔV requirement), in order to obtain the total mass of propellant needed, which amounts to 1.03 g.

Knowing the burning rate from Vieille's law and imposing a steady-state mass flow as in eq. 2.2, the area A_b of the end-burner is found to be a circle with a diameter of 0.5 mm.

$$A_b \rho_{prop} a P_C^n = \dot{m} \tag{2.2}$$

In order to fulfill the ΔV requirement given the low thrust of 5 mN the grain would need to be 3 m long, therefore a square array of thrusters was adopted instead of a single one. Since the maximum length-to-diameter ratio achievable by micro-machining is around 8, a combustion chamber length of 3 mm was selected, resulting in 1024 motors, arranged in a square configuration of 32 by 32, each producing 5 mN of thrust.

2.2.4 Array sizing

The spacing of the thrusters was computed to ensure mechanical resistance to the chamber pressure P_C. The associated wall thickness was calculated with the following formula:

$$t_w = \frac{k_s P_c d_c}{2\sigma} \tag{2.3}$$

Where k_s is a safety factor of 2, d_c is the diameter of the combustion chamber and σ is the yield strength of epoxy resin. This yields a wall thickness of 66.3 μm, which is very small compared to the diameter of the grain. Since it would probably not provide the required thermal insulation to avoid igniting nearby motors, an arbitrarily larger wall thickness of 0.5 mm was picked. From this, we find that the array has dimensions of 32.5 by 32.5 mm. Combining grain length, wall thickness, diverging nozzle length, as well as adding extra 5% to include some space before the throat, an overall array height of 3.86 mm is obtained, for a total volume of 4074 mm^3.

The total mass of the system is 7.97 grams, including the propellant. Below are reported the array's performance parameters

SRM Performance Parameters	
Parameter	Value
Thrust	5 mN
Mass Flow Rate	2.52 µg s^{-1}
Specific Impulse (I_s)	198.7 s

2.3 Monte Carlo Simulation

To evaluate the effects of manufacturing and propellant production uncertainties on the engine's performance, a Monte Carlo simulation was performed.
The parameters considered to be affected by uncertainty in the simulation include the Vieille's law coefficients, the characteristic velocity, the nozzle throat diameter, and the combustion chamber diameter. Their mean values and standard deviations are provided in Table 2.2.

As stated before, the uncertainties in the Vieille's law parameters "a" and "n" were derived from our previous analysis of BATES test motor firings.

Obtaining the uncertainties in the radii was more complicated as little information could be retrieved from the literature. For the combustion chamber radius, the uncertainty was derived from the tolerances of 0.5 mm end mills, as such a hole could be obtained by conventional milling techniques. The throat radius, on the other hand, requires a much more refined manufacturing process, and the only one able to produce a center-symmetric nozzle is FLM, as demonstrated

in [5]. Even still, it is hard to estimate the accuracy of such process when the throat requires to be only 33 μm across, and so an over-sized value for the error of roughly 1 μm was considered.

Table 2.2: Adopted uncertainties for SRM Monte Carlo simulation

Variable Name	**Symbol**	**Value (nom.)**	**Uncert. (%)**	**± Value**
Nozzle throat dia.	D_t	33 µm	3.0	1 µm
Combustion chamber diam.	D_c	500 µm	2.0	10 µm
Vieille's law factor	a	1.7270 mm/bar^{n}s	1.1	0.0184 mm/bar^{n}s
Vieille's law exponent	n	0.3821	0.7	0.0027
Characteristic velocity	c^*	1513.1 m s^{-1}	0.97	14.8 m s^{-1}

The Monte Carlo simulation generates N samples for each parameter, then picks N^3 out of the possible N^5 combinations of the 5 parameters to reduce computational cost. Every set of random variables is then fed to a function that computes the burning area, the throat area and then the chamber pressure using the following equation:

$$P_c = \left(\frac{A_c}{A_{tht}} \cdot c^* \cdot \rho_{\text{solid}} \cdot a \right)^{\frac{1}{1-n}} \tag{2.4}$$

From this, the performance of the motor is then computed. Table 2.3 shows the values of thrust, specific impulse and total impulse of both a single motor and the whole array that arises from the simulation, while figure 2.3 shows convergence of the numerical method.

Table 2.3: Output values from SRM Monte Carlo analysis.

Parameter	**Mean Value**	**Std.**
Thrust	5.02 mN	±0.19 mN
Specific Impulse	198.7 s	±1.9 s
Single Motor Total Impulse	2.02 mN s	±0.03 mN s
Array Total Impulse	2.068 N s	±0.001 N s

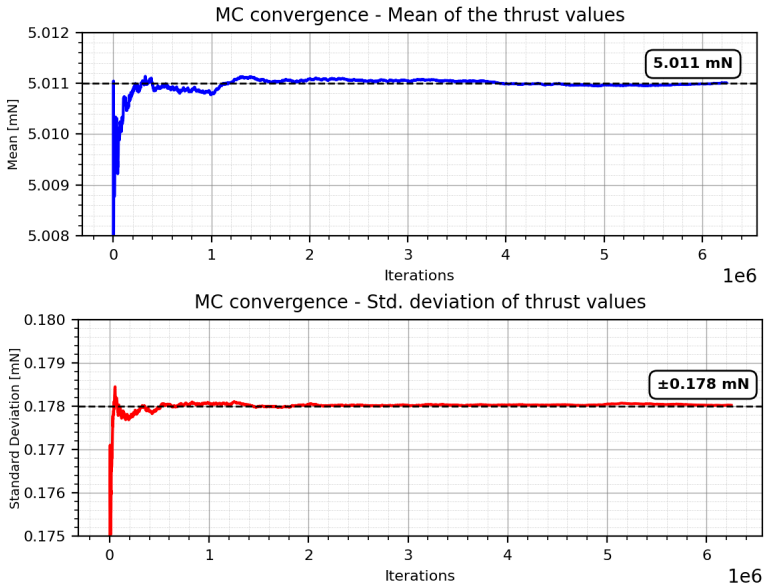


Figure 2.3: SRM thrust performance Monte Carlo analysis

3. Cold Gas Thruster

3.0.1 System Overview

Cold-Gas thrusters are one of the most simple and reliable reaction control systems in all space applications. The gas, stored at high pressure, passes through a nozzle and accelerates to high speeds, producing thrust. A simple design can be defined by considering an ideal isentropic nozzle. The main components of a typical cold-gas system consist of a tank, a valve, a pressure regulator, a plenum and a nozzle. The system must fulfill the following requirements:

1. **Pressure Regulation**: The design must guarantee the constant required pressure in the thrust chamber;
2. **Leakage Minimization**: The leakage related to each selected element must be considered.

The first requirement is achieved by implementing a pressure-regulated feeding system that operates until the expulsion of the nominal mass of fuel for the required ΔV. The leakage issue is addressed considering the leakage rate of the valves implemented in our system, and is managed by the pressurization system adding nitrogen mass to ensure the system's continuous work with pressure regulation.

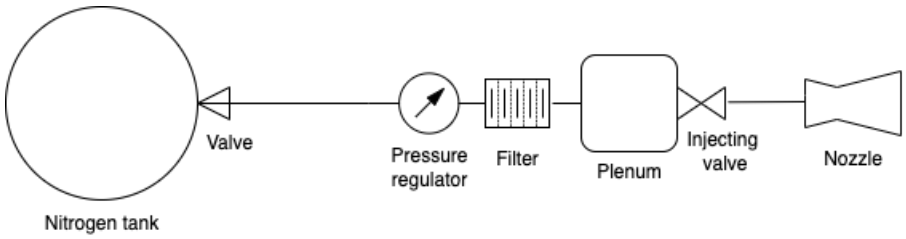


Figure 3.1: Selected design scheme.

Another important consideration regards the expansion ratio of the nozzle. During the expansion, the gas cools down, reaching low temperatures and risking a phase change in the divergent section, which, therefore, results in a liquid phase outlet. This issue imposes a constraint on the expansion ratio, in order to avoid gas condensation in the nozzle. This issue is closely related to the temperature of the gas in the inlet of the nozzle. An important assumption made throughout the entire sizing process is to consider the pressure losses as isothermal along the whole system. This assumption takes us to the more general one of constant temperature throughout the whole system. An exception is made for the tank, in which, considering a decrease of the temperature due to the expulsion of propellant, a resistor was implemented in order to maintain constant the temperature. The sizing also in this case was carried out considering as an objective obtaining $T = 5\,\mathrm{mN}$, this means that despite the losses, the value obtained at the end of the sizing must coincide with the one reported. The main source of losses considered in this study are the 2-D divergence nozzles in the nozzle, in order to simplify the approach, since it is the one only that can readily be computed analytically. However, at the present scale it is likely that losses due to boundary-layer effects might also be significant, however boundary layer modeling is highly complex and correlations for this factor in the literature are scarce and present a high range of variability. It is therefore to be expected that the real efficiency of the system will be lower than the one considered.

Among the main components of the propulsive system listed previously, some simplifications were made. In this sizing process, in fact, the presence of pressure and temperature sensors and most importantly filters was not considered. The main purpose of the filter is to block particles which can get stuck inside the valve causing leakage. A perforated silicon membrane is used to function as particle filter. The main constraint for the filter is the area of passage which should be much larger than the nozzle throat. However, due to the absence of pertinent literature, the calculation of the pressure losses introduced by those elements was omitted, considering them to be negligible.

3.0.2 Modeling approach

Once the general overview of the system is outlined, considering the simplifications and assumptions made, the sizing process can be described. The workflow that has been followed during this process can be initially summarized in the following figure (3.2) and successively in the next subsection analyzed in detail.

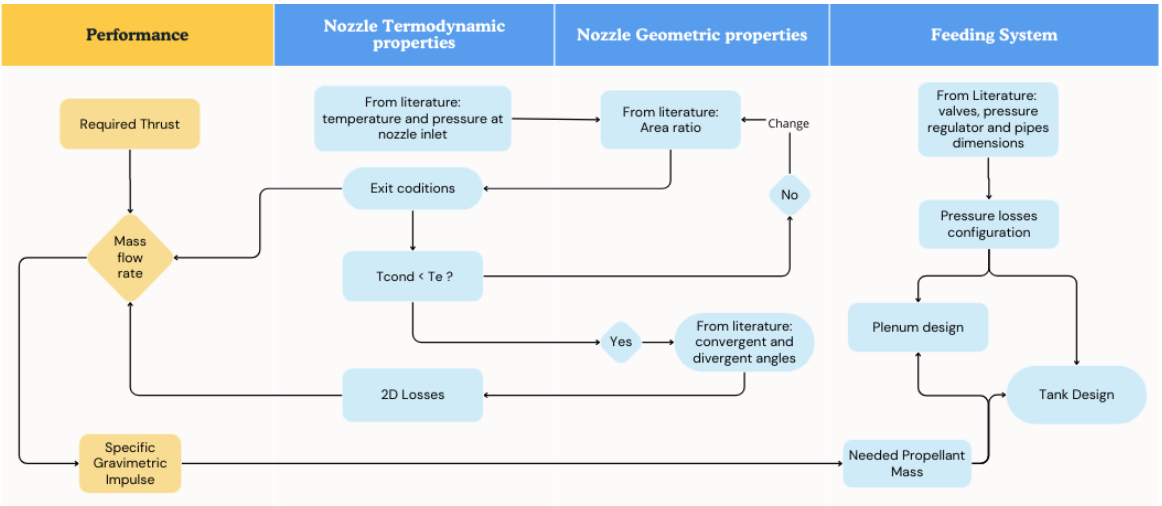


Figure 3.2: Adopted sizing workflow.

3.1 Nominal Design

3.1.1 Components

This section contains a more in-depth view of the components involved in the overall propulsion system in order to have a better understanding of the sizing process. The sizing was carried out in reverse order, starting from the nozzle, where the inlet temperature and pressure are fixed, and then working backwards until reaching the gas tank.

Nozzle

The high level model of the nozzle is based on an isentropic expansion and complete neglection of friction losses on the inside, while considering 2D losses. Two types of nozzles for MEMS use can be distinguished, the 2D extruded nozzle shape and the 3D conical nozzle shape. 2D extruded nozzles are usually made by means of deep reactive ion etching. The nozzles fabricated by this etch technique are limited in their geometry to extruded shapes and therefore the efficiency of these nozzles is low compared to the 3D conical variant. In our sizing we focus on the fabrication of a 3D conical converging-diverging nozzle. Considering literature given values, to obtain the required thrust the throat diameter should be in between 50 and 200μm, depending on the inlet pressure and expansion ratio. Both inlet pressure and expansion ratio together with inlet temperature are fixed in order to be able to carry out an appropriate sizing

process. In the same way, the angles of both converging and diverging sections of the nozzle have been retrieved from literature. The diverging part of the nozzle has a half-angle (α) of 28°. The angle of the converging part (β) of the nozzle is instead 45°, but is less important for the overall performance. The static thermodynamic properties at the inlet and the other imposed values are shown in the following table:

Inlet Temperature [K]	Inlet Pressure [atm]	Expansion Ratio [-]	Inlet Mach [-]
300	5	8	0.2

Table 3.1: Imposed values at nozzle inlet.

The selected material for the micro-nozzle is Silicon (Si) due to its efficiency for MEMS-scale nozzles and proven reliability in numerous real missions. Two potential technologies can be exploited to create a conical converging-diverging nozzle each of which has its own typical characteristics. The first method is deep reactive ion etching (DRIE). Consisting in controlling the profile of an etched hole to obtain the nozzle shape. For the second method, femtosecond laser machining (FLM) is examined. The energy in the laser spot is used to tune the angle of the nozzle sidewall. Among these two treatments, the one chosen for the fabrication of our nozzle is FLM because of its better precision regarding conical micro-nozzles made out of the selected material. The overall sizing of the nozzle has been computed through the same method that has been used for the other proposed propulsion systems, considering the previously discussed constrictions. From the sizing the subsequent values are obtained:

Section	Area [mm^2]	Diameter [mm]
Inlet	0.018	0.15
Throat	0.0062	0.089
Outlet	0.049	0.25

Table 3.2: Nozzle dimensions.

From the thermodynamic side the characteristics of the nozzle are:

Section	Pressure [bar]	Temperature [K]
Inlet	5.06	300
Throat	2.74	252
Outlet	0.053	81.6

Table 3.3: Nozzle thermodynamic characterization.

As the last parameter characterizing the nozzle we have the length, which, considering the previously given angles and the diameters calculated through the sizing, is set at $L = 0.18\,\text{mm}$.

Injection valve

The gas is inserted into the thrust chamber directly from the plenum, passing through a short pipe leading to an on-off valve. Valve characterization is obtained from the data belonging to a real valve datasheet. Passing through the valve and the pipe, a pressure drop occurs due to dynamic effect and orifice losses. The pressure loss design was carried out using the gas-dynamic model of a gas passing through an orifice described by the following formula:

$$Q = Y A_{valve} C \sqrt{\frac{2\Delta P}{\rho}} \tag{3.1}$$

In this equation [9], the known parameters were A_{valve}, ρ and c_d. The area of the valve was obtained from the diameter contained in the literature, $D_{valve} = 0.15\,\text{mm}$, and in the same way $c_d = 0.76$, which made possible the computation of the flow coefficient C:

$$C = \frac{C_d}{\sqrt{1-\beta^4}} \tag{3.2}$$

The value of the net expansion factor Y was computed using a linear approximation based on the graph for compressible flow through orifices presented in [10]. Expressing Y as function of $\frac{\Delta P}{\Delta P + P_c}$ it was then possible to apply an iterative method to the previously presented equation 3.1 in order to obtain the value of pressure losses across the valve $\Delta P = 0.39\,\text{bar}$.

Pipes

To connect the plenum to the nozzle, a pipe was modeled. The geometric characteristics of the pipes were assumed considering a diameter larger than the one of the injecting valve, $D_{pipe} = 0.4\,\text{mm}$ and a length compatible with the orders of magnitude of the whole system, $L_{pipe} = 1\,\text{mm}$. The dynamic losses in the pipes were then computed analyzing the Mach and Reynolds numbers of the gas ($M = 0.027$, $Re = 1324.1$) from which the the Darcy friction factor f was obtained as:

$$f = \frac{64}{Re} \tag{3.3}$$

as a consequence, the pressure losses in the pipes can be evaluated as follows:

$$\Delta P = \frac{1}{2}\rho v^2 \cdot f\frac{L}{D} \tag{3.4}$$

Considering the previously obtained Reynolds number and consequently the value of the Darcy friction factor $f = 0.0133$, the pressure loss across the pipe is $\Delta P = 34.51\,\text{Pa}$.

Plenum

The plenum was modeled as a volume needed to regulate and stabilize the mass flow before entering the injection valve section and to damp any possible upstream pressure fluctuations. Due to a lack of detailed of sizing information from literature, it is sized considering a portion of the total mass of propellant on board. In the considered study case the plenum was sized considering its content equivalent to 10% of the total mass of nitrogen evaluated from the Tsiolkowsky equation through given ΔV. Following this reasoning, the mass contained in the plenum is $m_{plenum} = 0.312\,\text{g}$ from which, through the density of the gas in that point, the volume of the component was evaluated, obtaining: $V_{plenum} = 50.95\,\text{cm}^3$. Additionally, the density was computed through the knowledge of the pressure in the plenum, also needed for the subsequent pressure regulator characterization. The pressure is computed considering the pressure losses encountered proceeding in reverse starting from the inlet nozzle condition, resulting in a value of $P_{plenum} = 5.46\,\text{bar}$.

Pressurization System

The pressure in the plenum is kept constant by an upstream pressure regulator. The pressure regulator works correctly if its upstream pressure is higher than the downstream wanted pressure, to avoid any malfunctioning, a limit was imposed on the minimum upstream pressure:

$$P_{MINup} = 1.5 \cdot P_{plenum} \tag{3.5}$$

In order to connect the pressure regulator to the tank another set of pipe and valve has to be sized, considering the valve immediately downstream of the tank and the pipe as a connection between the latter and the pressure regulator.

Upstream pipe

The reasoning on which the sizing is based is substantially identical to that proposed previously, with the exception of the dimensions of the pipe, which in this case are different due to the different positioning and linked components. In the proposed case the considered pipe is characterized by a diameter of $D_{pipe\,up.} = 0.11\,\text{mm}$ and a length of $L_{pipe\,up.} = 0.5\,\text{mm}$. The pressure losses can then be evaluated as before, obtaining a value of: $\Delta P = 0.02\,\text{bar}$.

Post-tank valve

Proceeding with the sizing, one of the last components is the valve immediately downstream of the tank, which has the task of regulating the mass flow rate coming out of the tank. The sizing has been carried out in the same way of the previously described valve, considering the pressure losses across an orifice. Also in this case the known parameters were A_{valve}, ρ and c_d. The valve is the same considered in the previous case, so both area and c_d have the same value, as well as C. Through an iterative method the pressure losses across the valve can be evaluated and, in this case, the value stands at $\Delta P = 0.62\,\text{bar}$.

Pressurized Tank

In order to determine the volume of the tank, we first need to consider how the pressurization works. To ensure that just before the end of operation in the tank there is still enough pressure to have an operating pressure regulator downstream ($P = P_{MINup}$) an additional mass of nitrogen was considered. Following this reasoning the total mass contained in the tank was increased considering a percentage of nitrogen dedicated to pressurization. Basing on the value of the total mass of fuel obtained from the Tsiolkowsky equation $m_{fuel} = 3.12\,\text{g}$, the needed pressurizing mass is:

$$m_{press} = 0.5 \cdot m_{fuel} = 1.56\,\text{g} \tag{3.6}$$

Considering this additional mass, the total mass contained in the tank will be $m_{tot} = 4.68\,\text{g}$. Another important consideration that must be addressed consists in considering a constant temperature throughout the system. To ensure that this hypothesis is reasonable, a consideration was made on the energy that a resistor has to provide to the gas in the tank to keep $T_{tank} = 300\,\text{K}$ while it empties. Considering these assumptions, only the sizing of the tank volume remains, obtained as:

$$V_{tank} = \frac{m_{fuel} + m_{press}}{\rho_{tank}} = 157.51\,\text{cm}^3 \qquad P_{tank} = \frac{1.5 \cdot m_{fuel} \cdot T \cdot R}{V_{tank}} = 26.49\,\text{bar} \tag{3.7}$$

Power Requirement

To compute the power needs of the systems in order to ensure the correct operation the following elements were considered: the power needed for the resistor inside the tank to keep the necessary temperature constant and, from literature, the peak power needed from the valve in switching between opened and closed configuration. Firstly, the power needed from the resistor is computed as follows:

$$P_{res} = c_v \, \dot{m} \, T_c \tag{3.8}$$

While from the datasheet of the considered valve the values of the used resistance, default and peak currents were obtained, from which the power consumption in both default and peak scenarios is evaluated, through the following equation:

$$P = R \cdot I^2 \tag{3.9}$$

The chosen value for sizing was the one referred to a peak condition, so as a consequence the needed power value per single valve is of $P_{valve} = 0.7\,\mathrm{W}$.

	Heating	Valves [x2]	Total
Power	1.65 W	0.7 W	3.05 W

Table 3.4: Propulsion System Power Requirements

3.1.2 Performance Results - Final System

Among the characterizing elements of the system, general performance parameters must be evaluated. At this point, after the sizing of the system, these parameters can be taken into account. A nitrogen-based cold gas Micro-Electro-Mechanical System (MEMS) thruster offers a compact, lightweight, and reliable propulsion solution for CubeSats. Typical performances include thrust levels in the millinewton range, with specific impulse (Isp) around 50–80 seconds. In this study also Total Impulse, Single Impulse Bit and Efficiency are considered. The Single Impulse Bit in particular was evaluated considering the minimum time spent by the injection vale in opening and closing the orifice, which vales is $\Delta t = 0.25\,\mathrm{ms}$. While in the evaluation of the efficiency of the nozzle only 2D losses were considered. The overall view of these parameters is reported in the following table:

CG Performance Parameters	
Parameter	Value
Power	3.05 W
Thrust	5 mN
Mass Flow Rate	$7.44\,\mathrm{mg\,s^{-1}}$
Specific Impulse (I_s)	68.53 s
Total Impulse	$2.10\,\mathrm{N \cdot s}$
Single Impulse Bit (I_b)	$1.25\,\mu\mathrm{N \cdot s}$
Nozzle Efficiency (η_{I_s})	94.4 %

3.2 Monte Carlo analysis

A stochastic simulation of the system was performed in order to evaluate the parameters sensitivity and to define the performance of the system considering the effect of uncertainties.

3.2.1 Uncertainties Modeling

The uncertainties relevant to the considered model are related to the orifice and nozzle manufacturing techniques. The first two parameters considered were the throat and exit diameters of the nozzle, which directly affect the expansion ratio and, therefore, the performances of our nozzle. The last parameter is the diameter of the injector that influences the mass flow rate. From the literature review of micro-nozzles manufacturing methods, a standard deviation of 5% and 10% were adopted for the nozzle throat and exit diameters respectively, based on previous work done on nozzles manufactured using femtosecond laser machining (FLM). It is worth noting that the presented approach allows for the expansion ratio to vary as a consequence of

the diameter uncertainties, which is an existent concern in the system design. Analogously, a similar approach was adopted for the uncertainty used for the injector diameter, where a value of 5% was adopted.

The uncertainties are all considered aleatory (as opposed to epistemic) since they are inherent to characteristics of the system (e.g. manufacturing methods), but well documented, and therefore modeled as Gaussian distribution with the uncertainties as standard deviations. A summary of the values adopted for the system's uncertainties is shown in the table 3.5:

Variable Name	Symbol	Value (nom.)	Uncert. (%)	± Value
Nozzle throat diam.	D_t	89.1 µm	5.0	4.45 µm
Nozzle exit diam.	D_e	252 µm	10.0	25.2 µm
Injector orifice diam.	D_{inj}	150 µm	5.0	7.5 µm

Table 3.5: Adopted uncertainties for Monte Carlo CG simulation variables

3.2.2 Stochastic Simulation Setup and Results

The Monte Carlo simulation consisted on the generation of a population with $N = 50$ values sampled from a normal distribution for each parameters. These arrays were shuffled with each other generating a total of N^3 combinations. The combinations were then randomized in order and the mean and standard deviation of desired outputs plotted as the population size grew. The simulation is allowed to be stopped early if an adequate convergence is verified for both metrics. The main parameter of interest, whose mean and standard deviation was utilized to evaluate the performance of the system, was the thrust, while also Specific Impulse and Total Impulse were evaluated.

The simulation is based on the evaluation of the thrust as:

$$T = \lambda\, u_e\, \dot{m} + P_C\, r_P\, A_e \tag{3.10}$$

where r_p is the pressure ratio.

The uncertainties are included in the evaluation of the thrust through a series of equations involving A_e, A_t and A_{val}. For instance, the value of P_c is obtained thorough an iterative method involving the mass flow rate through the injection valve equation 3.1 and r_p is obtained through an iterative method as well, involving this time the area ratio equation. Through this mechanism all the combinations of uncertainties are accounted for in the evaluation of the selected parameters, from which the mean value and standard deviation can be derived.

The results from the Monte Carlo analysis for this thrust is presented in figure 4.4. It was noticed that after about 125000 iterations, the solution stability was satisfactory, with oscillation levels well below the associated significant figures. It was noticed that the average thrust value, for example, was slightly lower than the one of the nominal design, which was within expectations given the non-linearity of the system. The nominal design operates at a point of optimal performance, and therefore displacements in any direction will lead to a reduced performance.
From 3.3, we can obtain valuable information that allows to understand how the propagation of input uncertainties affects the central tendency of the final thrust of the system. Also, as previously stated, specific and total impulse were evaluated through the Monte Carlo analysis, obtaining for each mean value and standard deviation. This allows for a better estimation of the system performance, as well as to express the result with more reliability and appropriate significance:

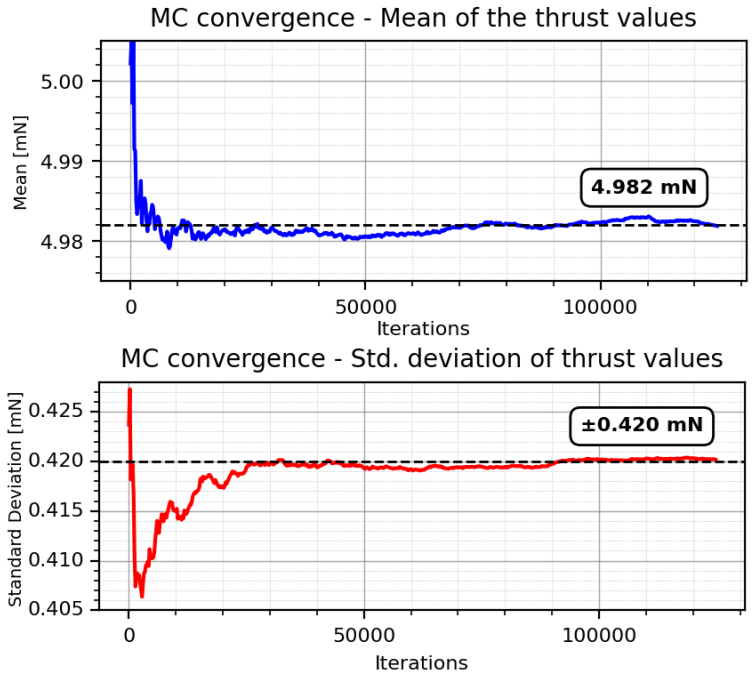


Figure 3.3: CG thrust performance Monte Carlo analysis.

Parameter	Mean Value	Std.
Thrust	4.98 mN	±0.42 mN
Specific Impulse	68.6 s	±0.8 s
Total Impulse	2.10 N s	±0.02 N s

Table 3.6: Output values from Monte Carlo analysis.

Statistically we observe that, for example, the average thrust is within 0.5% of the obtained nominal value, demonstrating the chosen design point is not highly sensitive. The standard deviation, on the other hand, is higher, but remains under 10% of the nominal value. This is deemed satisfactory considering the unavoidable high uncertainties propagated from nozzle manufacturing, and should be simple to adjust for during system testing.

3.3 Comment on Results

In conclusion of this study some important results should be pointed out. The whole sizing, as stated before, was based considering an increased mass flow rate in order to compensate the presence of 2D losses in the nozzle. Also several margins were applied to the total nitrogen mass present in the tank due to pressurizing needs and leakage issues. However, this design is not the global optima, but the simplifications made to the analysis in addition to the design assumptions made, led to this result.
The analysis can be further improved by taking into account several aspects, which include: accounting for the temperature change, more accurate pressurizing techniques and mass flow variations along with a more accurate description of the sources of performance losses.

4. Vaporizing Liquid Microthruster

4.1 System Overview

Vaporizing Liquid Microthrusters (VLMs) are one of the commonly used MEMS thrusters. They rely on the expansion of a heated gas or liquid to take advantage of its high energy to produce thrust as it exits through the nozzle. Even though the detailed design for different components is complex, the preliemary design can be reduced to a simple one-dimensional design, which gives an initial indicator on how the system will perform. The main components of the system typically consists of a propellant storage, feed system, with either a pressurizing gas or a pump, injector, vaporizing chamber (VC) and the nozzle. The VC design is one of the most critical steps in designing the VLM, since it contains the heaters responsible for vaporizing the propellant and also the volume to accommodate this transition. Through literature, it is pointed out that the VC design usually aimed to achieve the following requirements:

1. **Complete Vaporization**: The design must ensure complete vaporization of the propellant flow. [11].
2. **Performance Stability**: The vaporizer must be designed to manage the complex flow boiling patterns and minimize instabilities, as these directly influence specific impulse and thrust stability [12] [11].
3. **Heat Transfer Efficiency**: The design requires efficient thermal coupling between the heaters and the propellant to achieve the necessary vaporization with minimal power loss [11] [13].
4. **Performance**: The vaporizer design is critical for achieving target performance metrics (Isp, Thrust) by reliably producing vapor at the required state (pressure, temperature) before nozzle expansion [11] [12].

Regarding requirement 1, this is typically a matter of sizing the heaters and the power based on the required mass flow rate needed for the system performance. Requirement 2 also affects the dimensions of the VC and the placements of the heaters, which are either placed on the surfaces of the top and bottom wafers, or in the middle. Requirement 3 is achieved by using silicon in fabricating the VC wafers. this ensures that the maximum heat leak does not exceed 0.013 W/K [14].

The nozzle design is another critical aspect of designing the VLM in terms of the nozzle geometry and flow regimes encountered in it. The effect of boundary layer effects on propulsion efficiency was discussed in the introduction (1).

Another important consideration is the expansion ratio ϵ, as mentioned in (1). For high operating pressures, the performance increases with increasing ε. However, in the case of VLMs, increasing ε leads to faster supercooling of the expanded vapor, which can lead to formation of ice particles on the nozzle diverging part, which in turn reduces the performance through two-phase effects. To counteract this effect, the temperature of the superheated vapor needs to be increased further, consuming more power. This calls for an optimization that sets ε to maximize the performance in terms of power and expansion potential. Work in [11] performed this analysis and recommended an approximate value for ε of 7.

4.2 Nominal Design

The work flow used in the design of VLM is summarized in figure 4.1

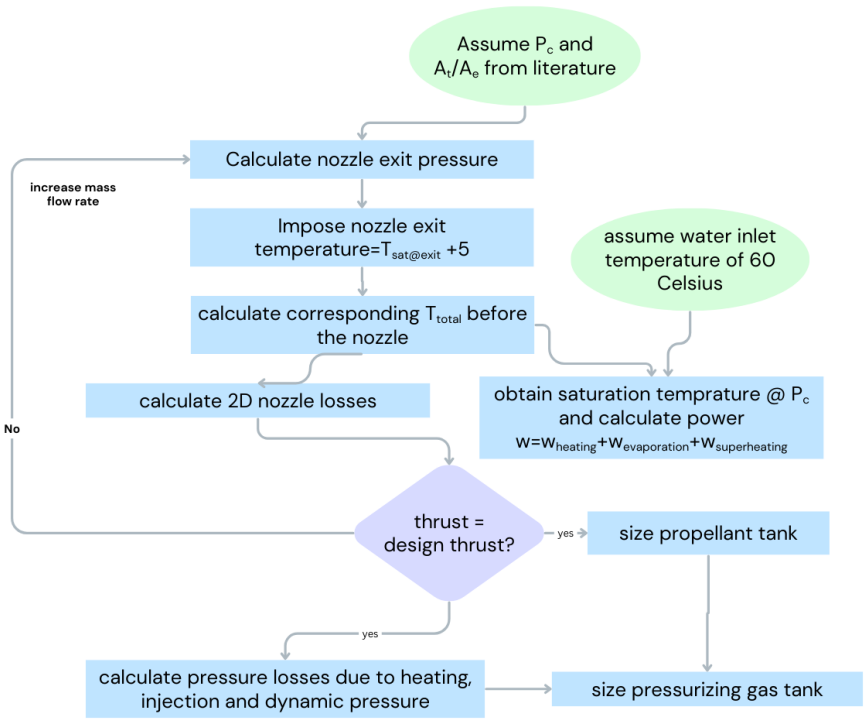


Figure 4.1: Work flow chart.

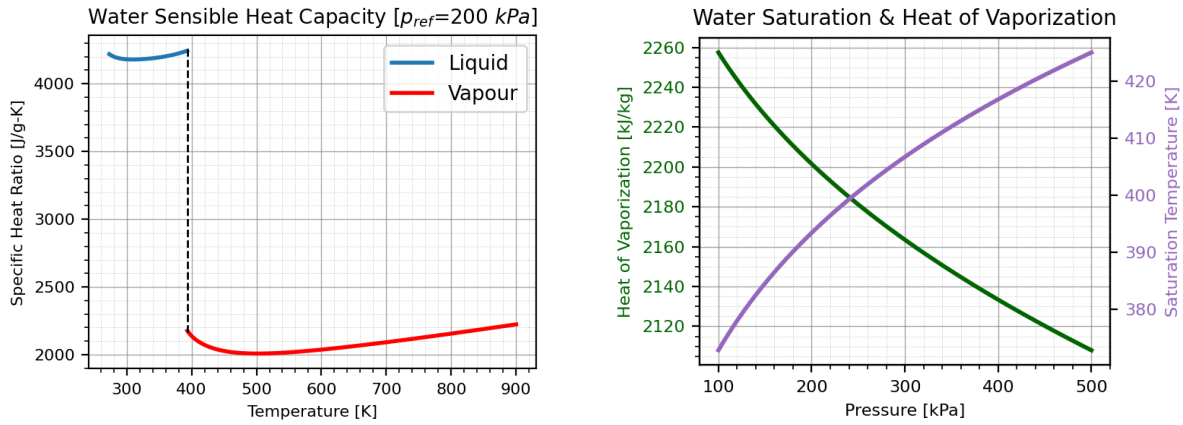
In the following table are reported some nominal performance metrics obtained by considering nominal parameters for the sizing of the system. The approach used to model and simulate the thruster will be described in the following sections.

Performance Metrics (Nominal)	
Parameter	Value
Nominal Thrust	5 mN
Nominal Specific Impulse (Isp)	155.1 s
Nominal Power Consumption	10.95 W
Nominal Mass Flow Rate	3.28 mg s^{-1}
Propellant Type	Water

4.3 Thermophysical Behaviour of Water

In order to properly estimate the heating power required for the vaporization of the water, it is necessary to estimate both the substance specific heat and the heat of vaporization. The heating process proceeds through 3 phases: sensible heat increase during liquid phase, vaporization heat to fully complete phase change, and finally sensible heat increase during vapor phase until the desired operating point. The model consists in a quasi-1D approach (divided into three regions), in order to incorporate heat transfer effects (Rayleigh approach) into the vapor flow inside the chamber.

The sensible enthalpy for each phase and the vaporization enthalpy will be modeled as a function of temperature, therefore the fluid is treated as non-calorically perfect. The values were retrieved for a given reference pressure in the vaporization chamber by means of the thermophysical library CoolProp [15], an open-source alternative to NIST databases. The behavior of the enthalpy is reported under figure 4.2:



(a) Water's $C_p(T)$ at 200 kPa

(b) Water's $h_{vap}(p)$ and saturation curve

Figure 4.2: Thermophysical behaviour of water sensible and vaporization heats

From this, the required enthalpy required to heat up a water mass unit to the desired vapor temperature for the thrust can be computed, as long as an operating pressure has been set. This procedure can be represented mathematically by:

$$\Delta h_{req.}(T_{des.}) = \int_{T_0}^{T_{sat}} Cp_{liq}(T)\,dT \;+\; h_{vaporiz.}(T_{sat}) + \int_{T_{sat}}^{T_{des.}} Cp_{vap}(T)\,dT \tag{4.1}$$

The required interpolation and integration routines were implemented, and therefore the required enthalpy to heat a fluid up until a desired temperature was mapped for different operating pressures and initial temperatures. Finally, with the required enthalpy rise the required power to be provided can simply be computed by:

$$W_{tot.} = \dot{m}_{H_2O} \cdot \Delta h_{req.} \tag{4.2}$$

Where the required mass flow rate is driven by the system thrust requirements. Note that the calculated power here consists of "effective" power actually delivered to the fluid. In the design of the heating apparatus of a real system, a certain thermal efficiency must be considered since the delivered power will be smaller than the total one due to losses, but this will be neglected in the present work.

4.4 System Components

4.4.1 Vaporizing Chamber

As explained in 4.1, sizing the VC is one of the most critical parts of designing the VLM. The work in [1] surveys a list of VLMs and their performances, based on the required nominal thrust of $5mN$, and values tabulated in this survey, a chamber pressure of 2 bar was selected to proceed with the analysis, as for the dimensions of the VC, we considered the work in [11] which used an operating pressure of 1 bar corresponding to a thrust of 1.01 mN, and used a VC dimensions of 7000∗3000∗100µm. Based on this, we set the dimension to be 9000∗3000∗100µm. The heaters could be placed in the middle of the chamber and equally distributed throughout its length. This would give a first-order estimate of the phase of the flow (or its quality in the case of two-phase) as a function of the position in the chamber:

$$\frac{L_{\text{sub-cooled-saturated water}}}{L_{\text{total}}} \approx \frac{w_1}{w_{\text{total}}} \;\middle|\; \frac{L_{\text{evaporating}}}{L_{\text{total}}} \approx \frac{w_2}{w_{\text{total}}} \;\middle|\; \frac{L_{\text{superheated steam}}}{L_{\text{total}}} \approx \frac{w_3}{w_{\text{total}}} \tag{4.3}$$

Rayleigh Flow

Pressure drop due to heat transfer effects were calculated for the superheated steam region, where density varies significantly with temperature. The analysis assumed a linear temperature rise with heat addition (proportional to distance) and used a quasi-fixed pressure (2 bar) for density calculations, justified by the small expected pressure variation. This resulted in a calculated stagnation pressure drop ΔP of approximately 3200 Pa (fig 4.3).

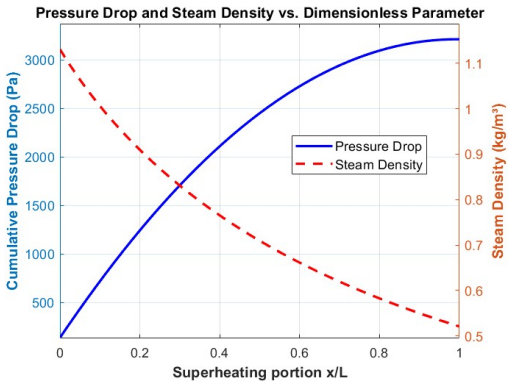


Figure 4.3: Pressure drop due to heat addition.

Plenum

The plenum (or inlet channel) ensures uninterrupted propellant injection into the VC at the required pressure, overcoming microscale challenges like high fluidic resistance. Following design criteria from [11], dimensions were chosen to minimize friction factor (f) by keeping the hydraulic diameter D_h above 500 µm. That work used a 1000 µm length and a 100x500 µm^2 cross-section. For this analysis, adapted for higher operating pressure, the plenum width was increased to 700 µm and length to 1200 µm. Located before the injector, the plenum must maintain a pressure equal to the VC pressure plus the injector pressure drop.

4.4.2 Heater Power and Energy

The function of the integrated heater in the micro-vaporizer system is central to its operation. The electrical energy supplied by the heater ($H = I^2R$) provides the necessary thermal power for the phase change process of the propellant.

Calculation Methodology

The power required to transform the propellant from its initial liquid state to a final superheated vapor state at the desired chamber conditions is calculated. This calculation is based on fundamental thermodynamic principles, described on the last section, and utilizes the previously calculated mass flow rate ($\dot{m}$), which is required to achieve the target thrust. The total power transferred to the propellant (W_{tot}) is determined by summing three distinct components:

- **Sensible Heat Power (w_1):**
  Power to heat liquid water from inlet temperature (T_{in}) to vaporization temperature (T_{evap}) at operating pressure.

$$w_1 = \dot{m}\bar{c}_{liq}(T_{\text{evap}} - T_{\text{in}}) \tag{4.4}$$

- **Latent Heat Power (w_2):**
  Power for phase change from liquid to saturated vapor at T_{evap}.

$$w_2 = \dot{m}h_{\text{vap}} \tag{4.5}$$

- **Superheat Power (w_3):**
  Power to heat saturated vapor from T_{evap} (enthalpy $h_{\text{gas,vap}}$) to the final stagnation temperature T_0 (enthalpy $h_{\text{gas,exit}}$).

$$w_3 = \dot{m}(h_{\text{gas,exit}} - h_{\text{gas,vap}}) \tag{4.6}$$

Calculated Heater Power Results

Table 4.1: Heat Power Components and Total Power

Component	Power (W)
Sensible Heat Power (w_1)	0.82
Latent Heat Power (w_2)	7.44
Superheat Power (w_3)	3.01
Total Power (W_{tot})	11.3

4.4.3 Nozzle

A key constraint is preventing water vapor condensation near the nozzle exit. The analysis first determines the nozzle exit pressure (P_e). Using an initial value for the chamber stagnation pressure ($P_0 = 2$ bar) and specific heat ratio ($\gamma = 1.32$), the isentropic flow relations for ϵ yield P_e.

From P_e, the corresponding saturation temperature (T_{sat}) is found. To provide a safety margin against condensation, the target exit temperature (T_e) is set 5 K higher than saturation ($T_e = T_{sat}(P_e) + 5$ K), resulting in $T_e \approx 302.2$ K. With (T_e) fixed, the required propellant stagnation temperature (T_0) is determined via isentropic relations. This T_0 dictates the total enthalpy change needed from the initial water state (60 °C) and thus the required heating power (w_{tot}). Following these considerations an expansion ratio of $\epsilon = 7$ was selected.

The analysis also accounts for 2D nozzle losses using the divergence factor (λ), calculated from the nozzle geometry. Work in the literature [11] recommends a semi-divergence angle of 28°, which will be used in this analysis as an initial estimate and changed accordingly. Using this value, λ is calculated to estimate the reduction in thrust compared to ideal axial flow, yielding a corrected thrust value.

For ease of analysis, an axisymmetric conical nozzle configuration will be considered. Note that for the case of the nozzles modeling in most literate work, which use rectangular nozzles. This remark can be improved in further analysis.

4.4.4 Injector Sizing and Flow Calculation

The flow of an incompressible fluid through a simple orifice, like the injector inlet, is typically modeled using the following equation, derived from Bernoulli's principle with corrections for real-world effects:

$$\dot{m} = C_d A_{\text{inj}} \sqrt{2\rho \Delta P_{\text{inj}}} \tag{4.7}$$

Where:

- $\dot{m}$: Mass flow rate of the propellant (kg/s). This is the target value determined from the overall thruster performance calculations;
- A_{inj}: Cross-sectional area of the injector orifice (m^2). This is the primary parameter to be determined by the sizing calculation;
- C_d: Discharge coefficient (dimensionless). This factor accounts for the difference between ideal theoretical flow and actual flow due to effects like friction and narrowing of the fluid stream after the orifice. It is typically less than 1. In this analysis, a value of $C_d = 0.70$ is assumed;
- ρ: Density of the liquid propellant (kg/m^3). For water, a value of $1000\,\text{kg/m}^3$ is used;

- ΔP_{inj}: Pressure drop across the injector (Pa). This is the difference between upstream fluid pressure and downstream chamber pressure (P_c). It is the driving force for flow through the injector. Pressure drop assumed as 15% of chamber pressure (Pc).

Rearranging the orifice flow equation to solve for A_{inj}. Once A_{inj} is calculated, the diameter could be found, assuming a circular orifice. Accurate sizing of the injector based on this flow equation is fundamental for micro-vaporizer performance:

- Undersized Injector: a smaller-than-required A_{inj} restricts flow, preventing achievement of target mass flow rate ($\dot{m}$) at designed pressure drop (ΔP_{inj}). This can result in insufficient propellant supply and lower thrust.
- Oversized Injector: a larger-than-required A_{inj} results in higher mass flow rates than required or lower-than-designed pressure drops. Low pressure drops can lead to unstable flow control and coupling with chamber pressure oscillations.

4.4.5 Propellant and Pressurizer Tank

The propellant and the pressurizer gas will be stored in a spherical tank with a bladder device in-between. Due to the low mass of the propellant relative to the CubeSat, the propellant mass can be estimated from a linearized version of the rocket equation: [14]:

$$\Delta V = g_0 I_{sp} \frac{M_p}{M} \tag{4.8}$$

The volume required for the propellant can then be found since the water density is known. Helium was selected as a pressurizing gas. We can size the pressurizing gas tank knowing that

$$V_{PG,final} = V_{PG,beginning} + V_{Propellant} \tag{4.9}$$

the following formula could be used to find $V_{PG,final}$:

$$V_{\text{PG,f}} = \frac{V_p}{1 - \left(\frac{P_{\text{PG,f}} \cdot T_{\text{PG,b}}}{P_{\text{pgb}} \cdot T_{\text{pgf}}}\right)} \tag{4.10}$$

As for the tank material, Aluminum AlSi10Mg was considered an option since it could be made using additive manufacturing and is also corrosion resistant, which is favorable in harsh space environment. The theoretical required thickness was much less than the precision current technology can achieve, so a thickness of 0.7 mm [16] was used to allow for a feasible solution.

4.4.6 Regulating valve

A valve is needed to regulate the flow in order to maintain the desired mass flow rate. Based on the maximum pressure difference between the pressurized gas and chamber, ΔP_{max} was found to be 2.3 bar. From this, the MEMTIS NC Classic valve was selected [17]. The datasheet provides the power required for different actuation currents, but an average, fixed value of 0.2 W will be assumed.

4.4.7 Pressure losses

Besides the pressure loss caused by the injection and heat addition, a distributed pressure loss of 100 Pa was assumed in the feed line, based on an approximated Darcy friction factor for flow in pipes. Additionally, with reference due to the use of 33 Gauge feeding needle in [18], a feeding line diameter of 110 µm was selected. This would result in a feed line velocity of 0.36 m/s, with a corresponding dynamic pressure loss of 63 Pa.

4.4.8 Final system dimensions and Performance Results

The following table summarizes the dimensions for different components and the performance results for the nominal design case. Note that only two-dimensional divergence was considered among nozzle losses.

VLM Design Parameters	
Nozzle	
Nozzle Throat Area	15 640 µm^2
Nozzle Converging length	320 µm
Nozzle Diverging length	400 µm
Nozzle length	720 µm
Nozzle semi-divergence angle α	20°
ϵ	7
Throat Knudsen number	9.9e-04
VC (Combustion Chamber)	
VC Dimension (L x W x H)	9000 x 3000 x 100µm^3
VC Operating Pressure	2.00 bar
VC Operating Temperature (outlet)	835 K
Power	11.5 W
Thrust	5 mN
Mass Flow Rate	3.38 mg s^{-1}
Specific Impulse (Isp)	150.8 s
I_{sp} efficiency ($\eta_{I_{sp}}$)	97.0 %
Impulse bit	4 mN s
Plenum Dimensions (L x W x H)	1200 x 700 x 100µm^3
Plenum Pressure	2.33 bar
Propellant	
Tank volume	5667 mm^3
Initial Tank Pressure	4.67 bar
Regulating valve volume	824 mm^3

4.5 Monte Carlo analysis

As for the other subsystems, a stochastic simulations of the system were performed in order to evaluate the parameter sensitivities and check for adequacy of the performance range the under the effect of uncertainties.

4.5.1 Modeling of Uncertainties

The uncertainties introduced in the model of the simulation system are related to the manufacturing capabilities of orifices/nozzles and to the provided heating power. The parameters which are treated as random variables are 1) Nozzle diameters (throat and exit, affecting ϵ), 2) Injector orifice diameter and 3) Heating power nominal value. From the literature review of micro-nozzles manufacturing methods, standard deviations of 5% and 10% were adopted for the nozzle throat and exit diameters, respectively, based on previous work done on nozzles manufactured by ion etching [19]. Due to the lower characteristic velocity of the working fluid, the subsequent obtained throat diameter was deemed suitable to be manufactured with such method. It is worth noticing that the presented approach allows for the ϵ to vary as a consequence of the diameter uncertainties, which is an existent concern in the system design. Analogously, a similar approach was adopted for the uncertainty adopted for the injector diameter, where a value of 5% was adopted. Finally, for the power, a breakdown of the resistor

heating element was done, in which the resistance was assumed to be under a nominal voltage of 5V, and the current is adjusted in order to provide the imposed nominal power. From the literature it was found that for space-grade components typical uncertainties lie on the order of 0.1% for the voltage and 0.3% for the current [20]. By Watt's law, this uncertainty can easily be propagated into the value for the power[21].

$$\sigma_W = \mu_W \sqrt{(\sigma_V/\mu_I)^2 + (\sigma_I/\mu_V)^2} \tag{4.11}$$

The uncertainties are all considered aleatory (as opposed to epistemic) since they are inherent to characteristics of the system (e.g. manufacturing methods), but well documented, and therefore modeled as Gaussian distribution with the uncertainties as standard deviations. A summary of the adopted values for the system's uncertainties is shown in table 4.2:

Table 4.2: Adopted uncertainties for Monte Carlo VLM simulation variables

Variable Name	**Symbol**	**Value (nom.)**	**Uncert. (%)**	**± Value**
Nozzle throat diam.	D_t	141 µm	5.0	7.05 µm
Nozzle exit diam.	D_e	373 µm	10.0	37.3 µm
Injector orifice diam.	D_{inj}	28 µm	5.0	1.4 µm
Heating Power	W	11.30 W	0.3	0.036 W

4.5.2 Stochastic Simulation Setup and Results

The procedure for the Monte Carlo simulation consisted on the generation of a population with N=30 values sampled from a normal distribution for each of the varying parameters. These arrays were then shuffled with each other generating a total of N^4 combinations, or 810000 maximum iterations. The combinations were then randomized in order and the mean and standard deviation of desired outputs plotted as the population size grew. The simulation is allowed to be stopped early if an adequate convergence is verified for both metrics.

The output parameter regarded as that of most relevance and whose mean and standard deviation was utilized to evaluate convergence was the delivered thrust of the VLM. The results from the Monte Carlo analysis for this thrust is presented in figure 4.4. It was noticed that after about 200000 iterations, the solution stability was satisfactory, with oscillation levels well below the associated significant figures, and therefore the simulation was stopped. It was noticed that the average thrust value was slightly lower than the one of the nominal design, which was within expectations given the non-linearity of the system. The nominal design operates at a point of optimal performance, and therefore displacements in any direction (in the 4-dimensional variable space) will lead to a reduced performance.

From the results in 4.4, we can obtain valuable information that allows to understand how the propagation of input uncertainties affects the central tendency of the final thrust of the system. This allows for a better estimation of the system performance, as well as to express the result with more reliability and appropriate significance:

$$T_{VLM} = 4.93 \pm 0.34\, mN \tag{4.12}$$

$$I_{sp,VLM} = 150.5 \pm 14.3\, s \tag{4.13}$$

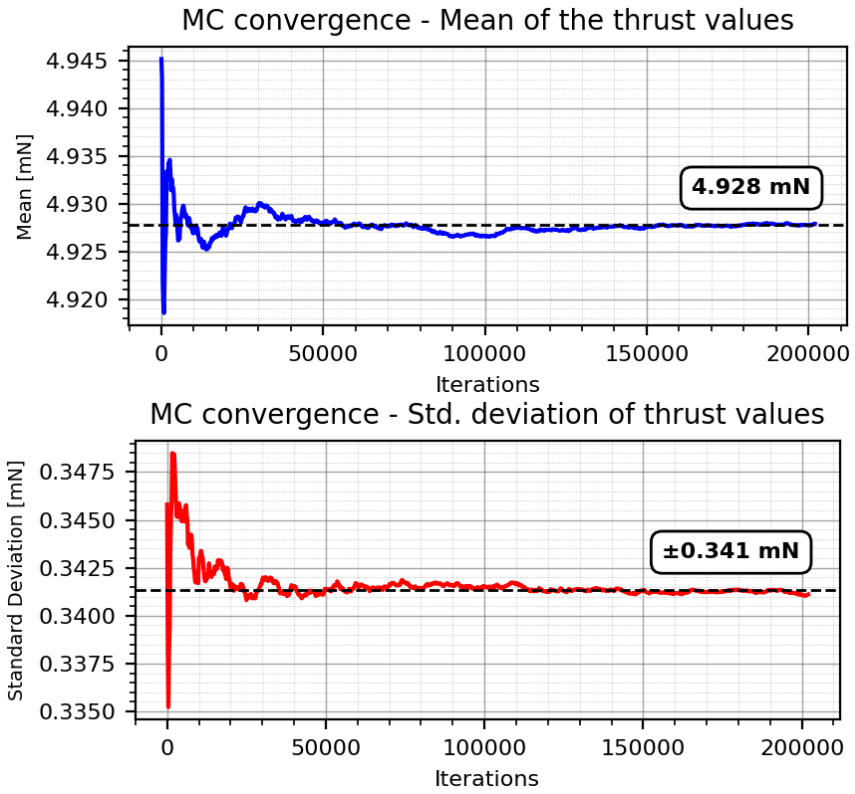


Figure 4.4: VLM thrust performance Monte Carlo analysis

Statistically we observe that the average thrust is within 2% of the obtained nominal value, demonstrating the chosen design point is not highly sensitive. The standard deviation, on the other hand, is considerably higher, but remains under 10% of the nominal value. This is deemed satisfactory considering the unavoidable high uncertainties propagated from nozzle manufacturing, and should be simple to adjust for during system testing.

4.6 Discussion of Results

A resistojet-type microthruster which managed to meet the imposed mission requirements was designed. This was done at a provided power level similar to those of similar-type propulsion systems found on the literature, but without considering any thermal efficiency for the heating process. The working temperature of the water vapor was brought to a level higher than what is typically seen in this type of system in order to ensure no two-phase flow at the nozzle's exit. However, it did not exceed the maximum operating temperature typical for many space-rated metal alloys, and therefore deemed not an issue.

The system propulsive efficiency was computed considering only divergence effects at the nozzle at losses, which is likely a significant gap in the modeling. Boundary layer effects on the nozzle performance are to be expected. Finally, the impulse bit was calculated using the time required to open and close the regulating valve plus the residence time in the VC.

In general, the analysis can be further improved by taking into account several aspects, which include but are not limited to the following:

- Accounting for power requirements related to thermal management, which is critical on in-space missions where water is required to be in liquid state.
- Implementing analytical methods that try to model microfluidic behavior of the propellant in such system, since viscous effects will have a significant contribution to the losses.
- Optimizing on operating temperature and expansion ratios to find an optimal trade-off.
- Considering reference fluids other than water, and evaluate their performances.

5. Comparative Analysis

In this paper, three different models for a propulsion system were proposed, each with its specific parameters and characteristics. In this last section, we aim to make a comparison to understand which technology is more suitable based on its specification.

5.1 Solid Propellant data analysis

A quick comparison between the designed engine performance and what is found in the literature points our design seems to stand out from the rest: in terms of specific impulse other motors reach a maximum of $\sim$ 200 s. The presented motor reaches a nominal value of 198.7 s. In terms of size, there is some significant difference: the smallest feature is about 100 µm in size, while the designed throat diameter was of 33 µm; and in terms of thrust others reach more than 1 N, while the current design is required to deliver 5 mN. These major differences can be attributed to two factors:

- Other motors are usually made just to test the feasibility of the system, rather than fully-fledged propulsion systems, and so often use low-performance propellant and/or use a nozzle with little to none expansion ratio.
- 5 mN of thrust is much lower than what is demanded by other motors, this combined with a high performance propellant and proper convergent-diverging nozzle results in much smaller systems.

Lastly the difficulty in estimating the uncertainties in the size of the nozzle due to manufacturing may have let to an underestimation of the actual uncertainties at play.

Parameter	**Thruster**	**Literature Range**
Thrust (F)	5.00 [mN]	500–2000 [mN]
Specific Impulse (I_s)	198.7 [s]	50–200 [s]
Smallest feature size (I_s)	33 [µm]	100–1000 [µm]

Table 5.1: Performance comparison of MEMS solid propellant thruster with literature data.

5.2 Cold Gas thruster data analysis

From the analysis conducted on a nitrogen-based cold gas MEMs emerges a good adherence with real values obtained from the literature. In fact, all the characteristic parameters obtained are reasonable with reference to real thrusters used in past missions. Analyzing the collected data it is possible to see how the 5 mN thrust output falls within the operational range of typical MEMS thrusters, while the mass flow rate of 7.44 mg/s aligns with high-pressure nitrogen systems (0.5-15 mg/s). At 3.05 W total power consumption (considering valve actuation and heating power), the system shows coherence with the implementation on a 3U CubeSat. The achieved specific impulse of 68.5 s (94.4% efficiency) exceeds typical values of nitrogen-based MEMS thrusters, this underlines the fact that there were no considered losses apart from the 2D losses, resulting in an exceptionally high efficiency, not coherent with real data.

With a single impulse bit of 1.25 µN $\cdot$ s, the system enables fine attitude control, meeting precision requirements similar to those of advanced missions. A cold gas driven thruster is vastly used for precision attitude control on CubeSats, due to its performances and characteristics such as: high reliability due to the absence of combustion, ignition or moving parts in the nozzle; high impulse bit, enabling fine attitude adjustments; re-usability and restart capability,

this thruster can be fired repeatedly, unlike solid propellant systems, making it excellent for long-duration missions or adaptive maneuvers. However, a downside of this design can be the occupied volume, due to the presence of a tank and a plenum that together occupy a volume of 208.5 cm^3.

5.2.1 Comparison with real data

The following table presents a comparison between the computed data from the sizing process with real data ranges obtained from literature. Overall the computed data are coherent with the ones obtained from past missions.

Parameter	Thruster	Literature Range
Thrust (F)	5.00 [mN]	0.10–20.00 [mN]
Mass flow rate ($\dot{m}$)	7.44 [$\mathrm{mg\,s^{-1}}$]	0.50–15.00 [$\mathrm{mg\,s^{-1}}$]
Specific Impulse (I_s)	68.5 [s]	50.00–80.00 [s]
Impulse Bit (I_b)	1.25 [µN s]	/
Power (P)	3.05 [W]	1.00–10.00[W]
Total Impulse (I_{tot})	2.10 [N s]	1.00–50.00 [N s]

Table 5.2: Performance comparison of MEMS cold gas thruster with literature data.

5.3 VLM data analysis

In general, the performance obtained from our analysis fall within the performance range encountered in the literature. The values, however, might be very optimistic since the analysis was carried out using the same analysis done in macro-fluids. The calculated specific impulse of 150.8 s is also within the broad literature range (6.9–180.00 s), situated towards the higher end.

This relatively high specific impulse value, which results in a calculated efficiency of 97%, could reflect the ideal assumptions made during the sizing process, where primarily nozzle divergence losses were considered, potentially underestimating other factors such as viscous effects within the nozzle or incomplete vaporization/thermal losses compared to values derived from the experimental literature.

Additionally, the power for the higher end of performance was not reported in the literature. The impulse bit appears large compared to cold gas thruster, which is mainly due to the residence time required to fully vaporize the water before expanding through the nozzle.

Parameter	Our Thruster	Literature Range
Thrust (F)	5.00 [mN]	0.006–6.5 [mN]
Mass flow rate ($\dot{m}$)	3.38 [$\mathrm{mg\,s^{-1}}$]	0.8–12 [$\mathrm{mg\,s^{-1}}$]
Specific Impulse (I_s)	150.77 [s]	6.9–180.00 [s]
Impulse Bit (I_b)	4[mN s]	/
Power (P)	11.5 [W]	1.00–10.80[W]
Total Impulse (I_{tot})	2 [N s]	/

Table 5.3: Performance comparison of MEMS VLM thruster with literature data. Note: not all research papers found had reported all of their parameters

5.4 Conclusion

In the following table is reported a comparison among the most characterizing performance values for each of the studied configurations. Now, with a complete view of each system specifications we can evaluate which system is more or less convenient in different scenarios.

	Solid	CG	VLM
Power	/	3.05 W	11.5 W
Specific Impulse	198.7 ± 1.92s	68.59 ± 0.75s	150.8 ± 14.3s
Total Impulse	2.068 N s (array)	2.19 N s	2 N s
Impulse Bit	2.02 mN s (single)	1.25 µN s	4 mN s
Actual Thrust	5.02 ± 0.19mN	4.98 ± 0.42mN	4.93 ± 0.34mN
Volume	4074 mm^3 (array)	157.51 cm^3+50.95 cm^3	5667 mm^3+2.70 mm^3

Table 5.4: Propulsion System Specifics Comparison

The volume considered as relevant for the three thrusters does not represent exactly the same elements, for the solid propulsion is considered the total volume of the array, for the Cold-Gas the main volumes are the Plenum and the tank ones, while for the VLM of the combustion chamber plus the tank one.

From these parameters, it can be seen that if a more precise control is needed, a cold-gas or a VLM can be a better choice. Moreover, while the cold-gas is less power-consuming, from a volume constraint point of view the VLM can be a more suitable solution in some applications. Meanwhile, if compactness and simplicity of the system are required, a solid propellant thruster might be a better choice.

The selection of the appropriate propulsion architecture is critical to achieve mission objectives within the restrictions imposed on mass, volume, and power. This report presented an in-depth study into the three most used micro-propulsion concepts — cold gas, solid propellant, and vaporizing liquid micro-thruster (VLM) systems — analyzing their performance considering fabrication driven uncertainties, and design challenges, proposing as a first step into a real life design of a MEMS.

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